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Sourcing and collecting data

Having decided what you want to do with data, it's now time to start thinking about sourcing and collecting it. For example, if you are using data to improve your decision-making, and you identified your key business questions as set out in Chapter 3, now you need to gather the data that will help you answer those questions.

Companies today have access to more data than at any other time in history. This should make data strategy a doddle – whatever you want to know, the information you need is going to be out there somewhere, right? Well, that's true, but finding it might not always be straightforward – and even when you do find it, getting it into shape so you can analyse it and extract insights presents many challenges.

A great deal of the data that companies have access to today is 'dark data'. This is a term used to describe information that we know exists, but we may not have the capability to catalogue and analyse. Sometimes it's dark because we don't know how to extract value from it – perhaps we are referring to a large archive of video footage, when we don't have the capability within our organization to digitize it and analyse it with computer vision. Sometimes it might be dark because it's locked away in physical archives, as might be the case with a legal or accountancy organization that has meticulously kept hard-copy files in its archives for decades. It could be social media posts made by our customers that could give us greater understanding of how our products are used and how we could engage with them more effectively – if only we had a way of decoding them.

The term 'dark data' derives from the term 'dark matter' used by physicists to identify the fact that we know something exists, but we don't know exactly what it is. We know this 'something' has an effect on the real world – on physics in the case of dark matter, and on our business and

operations in the case of dark business data. So, we build models that account for its existence to the best of our abilities – modelling the effect it has to the extent that we are able to. At the same time we try to improve our understanding of it, in the hope that at some point we will know enough about it to really understand the insights it can provide. Physicists do this by accelerating particles at close to light speed and monitoring how they behave under extreme conditions. Data scientists actually do something quite similar – throwing vast amounts of data into AI and machine learning algorithms to study it from every possible angle, with the hope of finding ones that throw light onto the dark data.

Iron Mountain, the 70-year-old data archival and data management company that we previously looked at, provides services aimed at helping companies get value from their ‘dark data’. Millions of square feet of storage space are taken up by paper documents and archives that their customers bring to them for safe keeping. But they also provide services around extracting and structuring the data in those documents for digital analysis. Texts can be scanned using computer vision and natural language algorithms that are smart enough to extract both content and context. This enables, for example, an insurance company to retain access to data on medical insurance claims stretching back years, in order to make more informed decisions today.

We’ve covered lots of examples in this book already of how various companies collect data, from Amazon tracking their customers’ purchases and combining it with data on how they browse their ecommerce portals, to John Deere aggregating datasets from its customers’ farms from sensors attached to tractors and harvesters. There are many ways to source and collect data, from plugging into the IoT to digitize and store real-world activities, to purchasing external datasets covering just about anything that can be imagined. In this chapter we will go through all of these methods. Remember, the smartest, most cutting-edge companies will automate these processes as far as is possible – data collection and entry is almost always a pretty dull job for a human, and these days there will almost always be a more efficient way of getting it done.

Before we jump in, it’s also important to remember that no one type of data is inherently better than any other kind. Using data strategically is about finding the data that works best for you and your use cases, and that’s likely to be very different from what’s best for another use case, or a different business. With so much data available these days, the trick is to hone in on finding the exact, specific data that holds the most value for what you are trying to achieve. This means that, from a data strategy point of view, you

need to describe the ideal datasets that help you achieve your strategic objectives. If you're lucky enough to find there's a range of different datasets that can fill this requirement, you can assess them based on how simple it is to access the data they contain and how cost-effective it is. You don't want to deploy a costly ML initiative to extract, say, customer behaviour data from a messy, unstructured dataset like social media posts or call centre logs, only to find that the same insights are sitting in far plainer sight in website click-stream data.

We take a deeper look at different types of data in the next section, but generally speaking, structured data is the easiest to analyse but not always the easiest to collect – because generally the world isn't a neatly structured place. At the other end of the scale, unstructured data exists everywhere – pictures, video, text and voice – so it can be collected quickly and cheaply but is more difficult (and expensive) to work with.

The more data, and the greater variety you have in your datasets, the fuller the picture that you can draw with them will be – and the more realistic your simulations and models that are trained on that data will be. In my experience it's often a combination of structured and unstructured, internal and external data that provides the most valuable insights. To meet your strategic goals, you may well need to mix some structured, internal data (such as sales records) with some structured external data (bought-in demographic data, perhaps), as well as some unstructured internal data (customer feedback transcripts) and unstructured external data (social media logs). It's when we start putting these disparate and self-contained datasets together that new connections and relationships between different elements of our business – customers, internal processes and so on – become apparent. This is where we're likely to find the most valuable insights.

Another way to think about it is like this – if the value in a dataset lies close to the surface and insights are easily obtained by simple analytical methods, it's unlikely we're going to be the first to have found it. It may help us find more efficient ways of doing something, but it probably won't give us a competitive edge.

Once you know what data you need, your next step is to identify how you will access and collect it. For internal data, collection methods include sensors, video, GPS, phone signals, IoT devices, network traffic, near-field communication (NFC) beacons, customer surveys and website user analytics platforms. For external data, we can look at social media analytics tools, as well as specialist tools that exist to access and crunch third party marketing databases, demographic data, electoral rolls, meteorological data, economic data,

news events and satellite imagery. Your choice of tools will depend on your strategic objectives, and we'll look at some of the most popular and effective options available later in the chapter.

You will also need to consider when you will collect the data. Is it data that we need to be gathering in real time so we can act on it as soon as possible? This might be the case if you're trying to identify potential customers passing your shop by their phone signal and send offers to tempt them in. Real-time data allows businesses to act on 'micro moments' – these are momentary opportunities to take action, such as when a potential customer arrives in the vicinity of your restaurant, at the time of day when they are most likely to be thinking about getting something to eat. Other data may only need to be collected daily, weekly or monthly – for example if you're collecting email addresses and using them to send targeted emails. There is no rule of thumb for when it's best to collect data; once again, you will need to be guided by your strategic objectives.

Understanding the different types of data

Data collection isn't new. Companies have had a lot of data for a long time (consider transactional records, HR files, newspaper archives and governmental records such as Hansard – the record of proceedings in the UK Parliament, or the US Library of Congress). Until recently, though, the only data we could really work with was *structured* data, meaning it was typically housed in spreadsheets or databases, which made it easy to interrogate. But advances such as the Internet, AI and cloud computing have increasingly made it possible for us to work with and extract value from unstructured data. One way to think about it is that more and more data that would previously have been considered 'dark data' is becoming available for capture and analysis, as analytic technology improves. Today, everyday activities like walking down a street, getting in a car or buying something in a shop all generate data that can be collected – our phones share signals with other devices as they pass them by, cars share data through GPS and data transmission to manufacturers or insurance companies, and transactional systems capture data on who is shopping and spending money. All of these sources of data can be tapped and used by companies to improve the way they do business. In the next part of this chapter we are going to cover some of the different categories that data can fall into. This is important because there are distinctions between them that make a difference to how they can

be collected and used. Understanding this will help you plan how to approach dealing with each type of data, once you've identified it as something you need.

Defining structured data

Structured data is any data or information that is located in a fixed field within a defined record or file, usually in databases or spreadsheets. Essentially, it is data that is organized in a predetermined way, usually in rows and columns. Structured data has a 'schema' – derived from the Greek word for 'shape' – that defines some rules or framework with which the data is organized. The basic concept of the schema is that it means any piece of data in the dataset can be directly compared with another piece. Structured data is commonly managed using Structured Query Language (SQL) – a programming language dating back to the 1970s which is used for querying data in relational database management systems. It doesn't use smart AI algorithms; it's based on simple 'if this, then that' logic of programmatic computation. For structured data, this is perfectly adequate, because all the information the code needs to understand the dataset is there in the schema.

The average business has the potential to tap into a vast amount of structured data. The most common examples include customer data, sales data, transactional records, financial data, number of website visits, and any kind of machinery datapoints (such as temperature logs in a refrigerated storage facility). It can also include things like mileage logs, quality control monitoring and inventory data. Today, structured data is the source of most analytically derived business insights – but that is changing.

Compared with the exciting world of unstructured data (which we'll get to next), structured data often gets a bad rap. I can see why. Despite being (for now) the most commonly used type of data, structured data represents just 20 per cent of all the data available in the world. The remaining 80 per cent of data out there is unstructured. As more of the world becomes covered by cameras and more of our conversations and activities take place online, we can expect that the proportion of unstructured data will increase even further. This is why it's widely understood that unstructured data is where a great deal of undiscovered value is waiting to be found.

Another downside is that structured data is less 'rich' than unstructured data. It offers us a more limited picture of what's going on. This means it's often wise to use other data sources alongside structured data, to enrich your insights and increase the chance that they will contain real, new value.

For example, structured data will tell you that hits on your website decreased 25 per cent last month. Which is great to know, but why? Social media chatter, on the other hand, might tell you that people are finding your site is very slow to load, causing them to head elsewhere. Those social media posts would be unstructured data. If you missed seeing them yourself, there wouldn't be a way for your systems to pick up on this valuable insight – unless you'd built them to be able to work with unstructured data.

On the upside, structured data has some big advantages: it's usually cheap to use, it's easy to store, and it's easy to analyse with simple tools. Of course, more advanced analytics tools are very useful if your structured data is truly vast in size or very fast-moving. Today's cutting-edge structured data-crunching applications are immensely powerful and impressive... Walmart's transactional and customer databases, for example, ingest more than 2.5 petabytes of structured data every hour. The company is able to combine this structured customer data (particularly who bought what, and when) with a variety of sources (like internal stock control records) to create sales promotions that are tailored to individual customers.

Even if, like most companies, you don't have 2.5 petabytes of structured data, your own structured data can still serve as an excellent starting point for gathering insights. That's why I think it's a mistake to ignore structured data altogether. It still has plenty to offer businesses – particularly when it's combined with unstructured data.

Defining unstructured and semi-structured data

Unstructured data is the term for any data that doesn't fit neatly into traditional structured formats or databases. Examples include email conversations, website text, social media posts, video content, photos and audio recordings. As you can tell, it's often text-heavy, but may also contain data such as dates and numbers, or other types of data such as images. Up until relatively recently, everything that didn't fit into databases or spreadsheets was usually either discarded or stored on paper or microfiche or scanned files that could not be easily analysed. Now, thanks to massive increases in storage capabilities and the ability to tag and categorize unstructured data, not to mention advances in analytical tools (more on that in Chapter 12), it is becoming increasingly possible to make use of this data.

It isn't quite accurate to say that AI and ML techniques like computer vision and natural language processing work with unstructured data. What they do is take unstructured data and convert it into structured data – by

using what they can learn about it to apply structure and schema. Once this has been done, it can be analysed and interpreted by traditional, programmatic computing methods.

Semi-structured data is a cross between unstructured and structured data. This is data that may have some structure that can be used for analysis (like tags or other types of markers) but lacks the strict structure found in databases or spreadsheets. For example, a tweet can be categorized by author, date, time, length and even the sentiment behind the tweet, but the content itself is generally unstructured. It is possible, these days, to automatically analyse the text in that tweet, but not using traditional analytical methods – I would need a specialist text analysis tool, such as one that can carry out sentiment analytics. This is a method of using NLP to determine the ‘mood’ behind the text – is it happy or sad, or was the author annoyed, concerned or amused when they created the tweet? A sentiment analytics algorithm can process the unstructured text in the tweet and add metadata that, as you can imagine, would provide valuable insight for anyone wanting to understand their customers’ behaviour on social media. In one example of sentiment analysis carried out on Twitter, researchers were able to predict which women were most at risk of developing postnatal depression. They analysed Twitter posts, searching for verbal clues in the weeks leading up to the birth. They found that negative language and words hinting at unhappiness, as well as an increased use of the word ‘I’, indicated an increased chance of developing postnatal depression. The end result of this would be posts tagged with an indicator of whether or not the poster was likely to be depressed – in other words, structure is added to the post as metadata.

You might guess from this that the main downside of working with messy, unstructured data is that it’s complex stuff, usually requiring specially designed software and systems. As a result, the costs can add up. Unstructured data tends to be much bigger than structured data, which means you need bigger and better storage, too.

None of this should put you off using unstructured data. I firmly believe that it’s where a lot of the undiscovered value that data has for businesses lies. It just means it’s important to be very clear about what you’re aiming to achieve and what data you need in order to do that.

Of course, the big advantage of unstructured and semi-structured data is that there is so much of it, and it is very highly descriptive and rich in content. A sentence constructed in natural language can convey all of the insight that’s important from a large structured dataset. For example, say

you want to know which driver has won the highest number of Formula One world championships. Starting from zero knowledge, to work this out from structured data, you would need to analyse the record of every driver who has ever won a world championship.

To work it out from unstructured data, you can simply put the question into a search engine like Google and it will scan news reports and other natural language sources to come up with the answer. The computation that goes on behind the scenes when we're using unstructured data is far more complex, but this is transparent to the data customer (us), who simply gets the answer presented in a clear and easy-to-understand way.

Here's another simple example of how we can now work with unstructured data more easily. Consider a video of a cat playing with a ball of string. A few years ago, for that video to be categorized (for example, so it would come up in search results), someone would have to watch it and tag it according to certain terms (cat, cute, ball, funny, etc) so that people searching for funny or cute cat videos could find it more easily. Now, videos can be automatically categorized using algorithms, meaning a computer can watch the video, automatically detect what's in it (maybe even who is in it, thanks to facial recognition software), and produce its own tags automatically. Brands are starting to use this technology as part of their everyday marketing activities. A friend of mine runs conferences for a living, and one of the conferences he ran was for a well-known electronics manufacturer. Just before the conference started, he shared a picture on Twitter of the main stage, ready for the first speaker. The picture featured the manufacturer's name and logo, which was on a sign behind the stage, but he didn't mention the company explicitly using a hashtag or their Twitter handle. So why, the week after the conference, did he keep seeing targeted ads online for that particular brand? Because the company knew he was talking about them; their analytical software was able to mine unstructured data like social media posts and photos for anything related to their company and their products.

Defining internal data

Internal data refers to all the information your business currently has or has the potential to collect by itself. Internal data can be structured in format (like a customer database or transactional records) or it could be unstructured (like conversational data from customer service calls or feedback from employee interviews). It is your private or proprietary data that is owned by

your business – and this means that only your company controls access to the data. There are many, many types of internal data, but some of the most common examples include sales data, financial data, HR data, customer service records, stock control data, CCTV video data, sensor data from company machines or vehicles, and your own website data (like number of visitors, click-through rates and so on).

One downside to internal data is the fact that you are responsible for maintaining and securing that data. This costs money and, particularly in the case of personal data, there are strict legal requirements that must be adhered to. When you buy in external data, on the other hand, the data supplier takes on that responsibility and liability for you. Another disadvantage is that internal data on its own may not provide enough information to meet your strategic goals, and you may need to supplement it with external data. Rather like blending structured and unstructured data to get a really rich picture of what's going on, often it is necessary to combine internal data with external data to get the most useful insights.

On the upside, internal data is usually cheap or free to access, which often makes it a good starting point when considering your data options. Plus, as you own the data, there are no access issues to deal with. You're never at the whim of a third party that can jack up prices or cut off access at any time. For really business-critical information, issues around access and ownership should not be taken lightly. Finally, there's real value in your internal data because it's already tailored to your business or industry. So, while you may need to look at some external data alongside your internal data to get the best results, you should never overlook it altogether.

One brilliant example of internal data is the Netflix example we looked at back in Chapter 4. As a content creator, Netflix's content strategy here has been firmly driven by its internal data – which is why it was able to confidently commission two series' worth of *House of Cards* episodes, as the show's format was so confidently predicted to be a hit by its algorithms, based on its internal customer data. Netflix's ability to mine their in-house data for valuable viewer insights really paid off.

Defining external data

External data is the infinite array of information that exists outside of your organization. This can be publicly available (like certain government data) or privately owned by a third party (like Amazon), and it can also be structured or unstructured in format. Key examples of external data include

social media data, Google Trends data, government census data, economic data, satellite imagery (unless you own the satellite) and weather data. There are plenty of ready-made datasets available, but if you need something more bespoke you can pay a third-party provider to gather the data for you.

The obvious downside to external data is that you don't own the data and you will often have to pay for access. This also means you are reliant on an external source, which can be risky if the data is absolutely critical to your key business functions. You will need to weigh up the risks and the costs of accessing external data against the risks and costs of not having that data. Would you have to go to the trouble of creating it yourself? Would your business suffer if you didn't use that data? Would it stop you meeting your strategic goals? You may find that, overall, the benefits far outweigh the risks.

There are some significant advantages to external data, though. Companies like Walmart and Amazon have the capabilities, infrastructure and budget to generate and manage huge amounts of internal data themselves. That's great for them. But many businesses can never dream of having that much data at their disposal. External data gives any business the capability to access and mine data for insights – without many of the hassles that come with storing, managing and securing that data on a daily basis. For smaller businesses, this can be a significant advantage.

Taking a look at newer types of data

The fact that we're leaving more and more digital traces than ever before means that many new types of data are popping into existence. Some of the data we can now collect is relatively new (like social media behavioural data), while some has been around for ages but we've only recently found ways to really analyse it (conversation data from customer service calls, for instance).

So I wanted to spend a bit of time highlighting the many new types of data that companies have at their disposal: activity data, conversation data, photo and video data, location data, satellite imagery and sensor data. It's important to be clear that all of these still fit into the category of either structured, unstructured or semi-structured data, and most can either exist as internal or external data. I have simply grouped them together here because they represent some of the biggest business-related leaps in data and analytics – which makes them useful considerations for any data strategy.

Activity data

This is the computer record of human actions or activities that occur online or in the offline physical world. If I think about everything I have done today before sitting down to write this chapter, most of those activities have left some digital trace that can be collected and analysed. My phone calls create data and, depending on who I speak to, the actual content of that call may be recorded and analysed – in the case of a call to a bank or customer service department, for example. Buying my wife a birthday present creates transaction data. Even browsing online for gift ideas creates a whole trail of data, including where I accessed the Internet from, what sites I visited, how I moved around those sites, what products held my attention and how long I spent on the sites. Everything I like on Facebook or share on LinkedIn or Twitter creates a trail. Even if I choose to switch off my phone and laptop and go for a run, my fitness band tracks my movements, how far I travel and how many calories I burned. Local CCTV cameras would also pick up my image along my favourite route. If I'm wearing a latest-generation Apple Watch, it will even be tracking the electrical signals passing through my body that keep my heart beating.

As you might imagine, the sheer volume of activity data available can make it difficult to pinpoint exactly what to collect. Continually circling back to your strategic objectives will help you concentrate on the best activity data for you, but it is hard not to get swept up in the many tempting possibilities that activity data presents.

On the positive side, activity data allows you to see what your customers actually do, as opposed to what they say they do or what you assume they do. This can be vital information for product or service development. Because we're creating more and more data every day, with practically every activity, there is an almost endless supply of rich data to tap into. Best of all, activity data is usually self-generating, which minimizes the amount of work for your business.

Conversation data

This doesn't just apply to an out-loud conversation with someone on the end of the phone. Conversation data also covers any conversation you may have in any format, from an SMS message or instant message through your phone, to emails, blog comments, social media posts and more.

Conversation data can be extremely useful for businesses because it provides insights into how happy or otherwise your customers, clients, employees and suppliers are. Conversations can be mined for content (what is said) as well as context (how, when and where it's said), giving us the ability to carry out sentiment analysis. In other words, you can understand what is going on from the words used and the mood of the person engaged in the conversation. This means companies can now figure out how angry or irritated a customer or employee is, or even if they are telling the truth about something, just from the stress levels in that person's voice.

Obviously, if you are planning to record any conversation, you need to be aware of any legal ramifications in your country. Generally speaking, you can't record customers or employees just because you feel like it; what you're recording must be relevant to the business. Also, you need to inform the parties that they are being recorded so that they can opt out if they wish. In addition, keep in mind that conversation data is also unstructured, which can make it more difficult and expensive to analyse.

On the plus side, conversation data gives you real-time access to customers and an accurate insight into what those customers really think and feel about your brand, product and services.

Photo and video data

Our ever-increasing attachment to our smartphones, and the commonplace use of CCTV cameras (particularly in the UK), has resulted in an explosion of photo and video image data. In days gone by, companies may have recorded their retail or storage premises for security purposes, but the recordings were never stored long term. After a week or so, the tapes would be reused and new recordings would be made over the old ones. Now, some of the more data-savvy stores are keeping all the CCTV camera footage and analysing it to study how people walk through the shops, where they stop, what they look at and for how long – very useful behavioural data that can be used to optimize layout of retail space and boost sales. Photo and video data is also used in industrial and manufacturing processes, for quality control purposes as well as monitoring the performance of machines. And the autonomous vehicles and self-driving cars we hear so much about use video data for many different purposes, including navigating around hazards and monitoring the condition of the surfaces they are travelling on.

Another type of image data that it's becoming increasingly easy to mine value from is satellite image data. The billions of dollars of investment that

has been poured into launching new satellites means this type of data is more affordable than ever, and today it's being used in agriculture to study crop yields and by the construction industry to identify areas prone to flooding, sink holes and other hazards. In the UK, Geospatial Insight provides data on vehicle traffic and footfall around shopping centres using satellite imagery, which can be used to understand how customers are travelling around and engaging with facilities.

Photo and video data can require a lot of storage space, which can be expensive to store and manage. It's therefore important to make sure you have a defined and relevant business need for collecting and storing this type of data. However, if you're already collecting this data as a matter of routine (perhaps through security footage), finding better ways to use it may not be very expensive at all.

Sensor data

As we've seen throughout the book, a vast amount of data is being generated and transmitted from sensors built into products. Your smartphone alone contains a GPS sensor, an accelerometer sensor (which measures how quickly the phone is moving), a gyroscope (which measures orientation and rotates the screen), a proximity sensor (which measures how close you are to other people, locations or objects), an ambient sensor (for adjusting the backlight on your phone) and a Near Field Communication sensor (this is what allows you to make a payment by waving your phone over a payment machine).

One issue with sensor data is that it often lacks context in isolation, meaning it most likely needs to be combined with other datasets to get truly transformational results. However, on the plus side, sensor data is self-generating, which makes it very appealing indeed. Many devices, such as smartphones, contain ready-to-use sensors that can be used to your advantage (think of a delivery company using sensors in their drivers' phones to track delivery routes).

Gathering your internal data

Having identified the data you need, it makes sense to see if you're already sitting on some of that information, even if it isn't immediately obvious. Consider whether the data you need already exists internally, or whether

you have the capability to generate it yourselves, for example by collecting data from your systems, products, customers or employees. Nowadays, you can gather data from any of your apps, software, or indeed any digital process – meaning almost any aspect of running a business can be monitored and analysed.

Wherever you are currently having conversations, there is an opportunity to collect conversation data. If you operate a telephone sales department or customer service department where customers call in to purchase or follow up on order delivery, you could record those conversations and analyse the content and sentiment for useful insights. Text-based conversation data also exists in internal documents and emails, and the emails you receive from your customers.

You can create your own data by asking questions and capturing the answers, through surveys, focus groups, asking people to rate your products, or by capturing details when customers register for something. You can also run experiments to gather data, for example, by running a marketing campaign, observing the results and tweaking parameters based on the insights you get back.

Video and photo data can be obtained by simply starting to collect it using digital cameras. Transaction data provides another mine of information for companies. It shows you what your customers bought and when. Depending on what you measure, it can also show where the item was purchased, how the customer came across the product and whether they took advantage of a promotion. Even basic transaction records can be very useful for measuring sales, monitoring stock levels and predicting what you need to order (or manufacture). In fact, all your company's financial data, not just the transactions, should be considered. Finance data has many uses, such as predicting cash flow and influencing investment and other long-term business decisions. It can be especially powerful when combined with other kinds of data. For example, you might look at your own internal financial data along with external data about industry trends and the wider economy. This is likely to give you insights into how external factors that are beyond your control will have an impact on your bottom line, helping you to plan and forecast more efficiently.

Crucially, you can use your smart products and services, as discussed in Chapters 3 and 4, to gather data. Sensor data is particularly helpful in this respect, and, these days, sensors can be incorporated into almost anything – from manufacturing equipment to shop doors to tennis racquets. Sensors are tiny, affordable and very easy to add to products. Swedish car

manufacturer Volvo, for example, is using data to improve driver and passenger convenience and create more user-friendly vehicles. Volvo monitors the use of applications and comfort features to see what their customers are finding useful, and what is being underused or ignored. This includes entertainment features like built-in connectivity with streaming media services, as well as practical tools such as GPS, traffic incident reporting, parking space location and weather information.

It's clear that internal data can be a gold mine and is an essential part of any good data strategy. Even if you need to combine your internal data with some external data to get a fuller picture, the data you already have (or have the potential to capture) is so unique to your business, it should never be overlooked.

Accessing external data

The most effective and revolutionary data initiatives tend to work by combining newly created internal data with external data that's already out there. As more and more companies view data as a business commodity, a market is emerging where practically any organization can buy, sell and trade data. (Indeed, many companies exist purely to supply other companies with data.)

Experian, as we've mentioned previously, is one example of a company that has diversified from collecting data for one purpose into selling it for any number of purposes. Another is Nielsen, which has existed since 1923 and pioneered many methods of market research and audience measurement in the US during the 20th century. Today it makes data available as a service on audience and consumer habits, to help other companies make data-driven decisions.

There are also lots of specialized, industry-focused data providers. So even if you need quite specific data, there's a good chance someone else is already gathering it. For example, Corelogic is a company that focuses on providing data and analytics services for the mortgage industry; it holds information on over 795 million historic property transactions and 93 million mortgage applications.

External data doesn't have to be expensive – in fact, plenty of it is available completely free of charge. Lots of information about healthcare around the world is collected and made available by the World Health Organization, for example, while organizations like the International Monetary Fund and World Economic Forum collate data on economic issues.

In addition, a lot of valuable data is being collected and shared by open government data initiatives, scientific research organizations and other not-for-profit agencies. Most governments these days try to make as much of their data as possible available free of charge. This can be a great source of information on everything from population to weather and crime statistics. The US government makes large volumes of data available on its data.gov site, and the UK equivalent is data.gov.uk.

Many countries also conduct a national census, which can be a very useful source of population data, geographic data and education data. Demographic data like this can be a useful indicator of trends, which is especially helpful if you're developing a new product or service. It can also help you target products or services to particular local demographics.

Clearly social media platforms are critical sources of data, and they provide a wealth of information on customers. Unlike some of the sources mentioned above, they don't tend to offer direct access to the data they hold – clearly as a lot of it is personal data, they can't do this. Instead they offer tools and services that let us get insights from the data, without seeing the data itself. Facebook offers incredibly useful breakdowns of customer information and behaviour – some of which you will need to access through its paid services, but some is available for free. Recently Facebook has also added the ability for anyone to upload their own customer data, which it will combine with its own data, to find other people who fit similar profiles who might make good customers for you. This is a great example of internal data being combined with external data to deliver value that couldn't be found in either dataset alone.

Twitter is another excellent source of data. Every time a Twitter user mentions a company or product, it is visible to everyone, including the company. Even if a product isn't mentioned explicitly in the text of the tweet, companies can detect if their product features in a photo. Examples of this might include a drinks company finding pictures of people drinking their product, restaurants finding pictures taken in their restaurant or fashion houses finding out who is wearing their clothes. AI and machine learning tools exist to automate this process and convey insights to brands in real time.

Google Trends is a very powerful and versatile tool, providing statistics on search volume (as a proportion of total searches) for any given term since 2004. You can see search popularity for certain phrases or words and how it has changed over time, and you can narrow the results by your geographic location. This is helpful for understanding trends in your industry, what is popular right now and what is becoming more popular (or less popular). It's a great way of gauging consumer interest.

Weather data, like that available through the US National Climatic Data Centre or the UK's Met Office, can be used in a number of ways, from estimating customer numbers and planning staffing levels, to deciding when to clear away the winter stock and bring out the barbecues and beach umbrellas. Lots of this data is available in the form of satellite images, but satellite images are another form of external data that have a great number of possible uses – as covered in the section above on image data.

There are thousands upon thousands of options for accessing external sources of data, and the options are growing every day. Keep in mind, though, that you're looking for the right data for you – and that means data that helps you achieve strategic goals.

When the data you want doesn't exist

When the data you need doesn't already exist, you have to find ways to generate it yourself. In many sectors and industries, businesses are fighting to be the first to collect new data and convert it into value. Often there is a distinct competitive advantage in being the first company to collect a certain dataset – for example Uber has become hugely successful by understanding how people travel around cities and learning how to have cars available at the time and place they are needed.

One way to create the data you need when it doesn't exist is to create products and services that are designed to capture it. Many of the products and services we covered in Chapters 3 and 4 would fit this bill – for example Google's Nest thermostats that 'learn' how customers use energy in their homes, smartphone manufacturers like Apple and Samsung that monitor how their products are used, and John Deere's smart agricultural machinery that collects data on how farmers work and how crops grow. All of these are constantly creating new datasets that give the manufacturers and service providers data that they can use to refine their offerings.

Another agricultural service provider, Springg, has come up with a way of collecting and analysing data on the fly, and then making it available to farmers in developing countries. The company recognized that farmers in developing nations could benefit from the same data as that available to farmers in developed nations, such as soil quality data, but don't have such easy access to the facilities needed. For example, in developing countries, the practice of taking a soil sample and then sending it off to a lab for analysis is far more expensive and takes far longer than in developed countries, meaning insights are far harder to obtain. Springg developed mobile test

centres with IoT devices that test soil remotely, give results almost immediately, and then send the data back to a central repository for further analysis. Obviously, this information benefits the farmers involved, but it's also a big win for Springg, which has collected aggregated data on soil conditions from places where it has never been done before. Finding innovative ways of collecting new data like this helps companies get a valuable first-mover advantage. This is true in more or less any area of data and analytics. Weather companies, for example, are constantly competing to be first to develop new ways of collecting data.

One final, but fascinating, area of data creation we will take a look at is what is known as synthetic data. Simply put, the idea here is that, when real data isn't available – perhaps it would be too expensive or intrusive to collect – why not simply create artificial data that mimics real data in every way, which can then be used to train machines and AI algorithms in exactly the same way as real data can?

Synthetic data has a few advantages over real data. One is that it's comparatively cheap to 'collect' – in fact it isn't collected, it's generated by algorithms, but the end result is the same. Another advantage may at first seem somewhat counterintuitive. Synthetic can sometimes more accurately represent the real world, because it will have been 'collected' (generated) without being affected by bias that might be present in real-world data-gathering operations. A popular technique for doing this involves generative adversarial networks (GANs) that use the same algorithms as those used to create 'deepfake' images of celebrities. So for synthetic facial data, rather than attempting to mimic famous people, it can simply come up with thousands or millions of images of entirely made-up people who never existed. These images are just as useful for training facial recognition algorithms as pictures of real people are, but can be generated far more quickly than real facial images can be captured, without any of the privacy or consent implications.